

InterviewAce

Requirements Traceability Matrix

Test Cases ↔ Use Cases | Backend Scope

Document Type	Requirements Traceability Matrix
Platform	InterviewAce
Scope	Backend — FastAPI / AI Scoring Engine
Prepared By	Backend Engineer
Date	May 2026
Version	1.0

1. Purpose

This Traceability Matrix maps each Test Case in the InterviewAce Test Case Specification Document to one or more Use Cases from the Use Case Analysis. It ensures complete backend test coverage and demonstrates that every use case identified for the backend scope has at least one associated test case.

2. Use Case Reference Summary

The table below lists all use cases in scope for the InterviewAce backend, derived from the Use Case Analysis document. The Job Seeker is the primary actor for all use cases; the AI/ML System acts as a supporting secondary actor for scoring and feedback flows.

UC ID	Use Case	Actor	Priority
UC-AUTH	Authenticate User	Job Seeker	Critical
UC-SCORE	Score Answer (AI)	Job Seeker (AI/ML System: secondary)	Critical
UC-FEEDBACK	Generate Feedback	Job Seeker (AI/ML System: secondary)	High
UC-PROGRESS	Track Progress	Job Seeker	High
UC-QBANK	Manage Question Bank	Backend seed/admin process	High
UC-ROLE	Select Job Role	Job Seeker	Medium
UC-SCORE-VIEW	View Score & Feedback	Job Seeker	High
UC-PRACTICE	Practice Interview	Job Seeker	Medium

3. Traceability Matrix — Test Cases to Use Cases

A filled circle (●) indicates that the Test Case directly validates the corresponding Use Case. Each column represents a Use Case; each row represents a Test Case.

TC ID	Test Case Title	AUT H	SCO RE	FEE D	PRO G	QBA NK	ROL E	S-V	PRA C
TC-01	JWT User Authentication	●							
TC-02	Answer Submission & AI Scoring		●						
TC-03	Feedback Generation During Scoring		●	●				●	
TC-04	Answer History Retrieval				●				
TC-05	Role-Based Question Retrieval					●	●		●
TC-06	Answer & Score DB Persistence		●						
TC-07	Deterministic Numeric Scoring		●						
TC-08	JWT Protection on Protected Endpoints	●							

4. Coverage Summary

UC ID	Use Case Name	# TCs	Status
UC-AUTH	Authenticate User	2 (TC-01, TC-08)	✓ Covered
UC-SCORE	Score Answer (AI)	4 (TC-02, TC-03, TC-06, TC-07)	✓ Covered
UC-FEEDBACK	Generate Feedback	1 (TC-03)	✓ Covered
UC-PROGRESS	Track Progress	1 (TC-04)	✓ Covered
UC-QBANK	Manage Question Bank	1 (TC-05)	✓ Covered
UC-ROLE	Select Job Role	1 (TC-05)	✓ Covered
UC-SCORE-VIEW	View Score & Feedback	1 (TC-03)	✓ Covered
UC-PRACTICE	Practice Interview	1 (TC-05, indirect)	✓ Covered

Note on UC-PRACTICE: The 'Practice Interview' use case is an orchestrating workflow that invokes UC-ROLE (question retrieval) and UC-SCORE-VIEW (answer + score delivery). It is indirectly covered by TC-05 and TC-03 and does not require a dedicated backend test case, as its underlying sub-flows are independently tested.

5. Key Relationships

The Use Case diagram defines two mandatory «includes» relationships that must be enforced by the backend:

- **Submit Answer → Score Answer (AI):** TC-02 validates that a single POST /scoring/submit call always triggers the AI scoring pipeline — mandatory sub-flow.
- **Score Answer (AI) → Generate Feedback:** TC-03 confirms that feedback is generated and returned inline with the scoring result from POST /scoring/submit, with graceful degradation when Ollama is unavailable.